

横向知识点对比

各大运算的等价性质

$$\begin{aligned} A \text{可逆} &\Leftrightarrow A^{-1}, A^*, A^T \text{可逆} \\ A \text{正定} &\Leftrightarrow A^{-1}, A^T, A^* (|A| \neq 0), kA (k > 0), P^T A P \text{正定} (P \text{可逆}) \\ A \text{为正交矩阵} &\Leftrightarrow A^{-1}, A^*, A^T, -A \text{为正交矩阵} \end{aligned}$$

$$\begin{aligned} A \sim B &\Leftrightarrow A^{-1} \sim B^{-1} (|A| \neq 0 \text{时}), \quad A^T \sim B^T, \quad (\Rightarrow A^* \sim B^*, \text{当} |A| \neq 0 \text{时}) \\ A \simeq B &\Leftrightarrow A^{-1} \simeq B^{-1} (|A| \neq 0 \text{时}), \quad A^T \simeq B^T \quad (\Rightarrow A^* \simeq B^*) \end{aligned}$$

特殊情形

特殊矩阵1

$$\begin{aligned} A = \alpha\alpha^T + \beta\beta^T, \quad \alpha^T\beta = 0, \alpha\alpha^T = 2, \beta\beta^T = 1 \quad r(A) \leq 2 \\ A\alpha = 2\alpha, A\beta = \beta \end{aligned}$$

$$\begin{aligned} A = \alpha\beta^T + \beta\alpha^T, \quad \alpha^T\beta = 0, \alpha\alpha^T = 2, \beta\beta^T = 2 \quad r(A) \leq 2 \\ A\alpha = 2\beta, A\beta = 2\alpha \\ A(\alpha + \beta) = 2(\alpha + \beta), A(\alpha - \beta) = -2(\alpha - \beta) \end{aligned}$$

注：A均为实对称矩阵

特殊矩阵2

$$A = E - axx^T \rightarrow A^2 = (E - axx^T)(E - axx^T) \quad (\text{寻找矩阵关系})$$

待整理问题

证明题：相似，正定等

结论证明

$$\begin{aligned} r(A) + r(B) - n &\leq r(AB) \\ A_{m \times n} B_{n \times s} \quad \left(\begin{array}{cc} AB & O \\ O & E_n \end{array} \right) &\rightarrow \left(\begin{array}{cc} AB & A \\ O & E_n \end{array} \right) \rightarrow \left(\begin{array}{cc} O & A \\ -B & E_n \end{array} \right) \quad (\text{分块矩阵初等变换}) \\ r(AB) + n &\geq r(A) + r(B) \\ \text{即} r(A) + r(B) - n &\leq r(AB) \end{aligned}$$

计算严谨性问题

经验表明，矩阵运算的出错率最高，弱智错误；勿急，一步错步步错